

ExamA.pdf



Anónimo



Automatas y Matematicas Discretas



1º Grado en Ingeniería Informática del Software



**Escuela de Ingeniería Informática
Universidad de Oviedo**



MÁSTER EN

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¡PARTICIPA EN NUESTRO **SORTEO EXCLUSIVO** **PARA ESTUDIANTES!**

Gana uno de los 3 portátiles ASUS Vivobook S15



April 3, 2014

- Let $G_1 = (\{S, A, B\}, \{a, b\}, S, P)$ be a CFG, where P is:
 $S \rightarrow AaB \mid ab$
 $A \rightarrow bA \mid b$
 $B \rightarrow bB \mid aB \mid a \mid b$
 Using the algorithm learnt in class, write a new CFG $G_1' = (V', T, S, P')$ in CNF. Then,
 - V' will contain 6 variables
 - V' will contain 5 variables
 - V' will contain 3 variables
 - None of the above.
- Consider a PDA where $\delta(q, a, A) = \{(p, BB), (q, AA)\}$. Then it holds that:
 - $(q, aaa, A) \vdash (q, aa, AA) \vdash (p, a, BB)$
 - $(q, aaa, A) \vdash (q, aa, AA) \vdash (p, a, BBA)$
 - $(q, aaa, A) \vdash (q, aa, AA) \vdash (p, a, BA)$
 - None of the above.
- After applying CYK algorithm to a certain grammar $G = (V, T, S, P)$ and string aab , $V_{1,3} = \{A\}$ (cell column 1, row 3) and $A \neq S$. Then, it holds that:
 - If $S \rightarrow AA$ is a rule in P, then the string $abaaab$ is in the language generated by G .
 - If $S \rightarrow SS$ is a rule in P, then $abaaab$ is a string in the language generated by the grammar
 - The string $abaaab$ is not in the language generated by the grammar
 - None of the above.
- Let G be a CFG and $w \in L(G)$. If there are two different parse trees for w , then:
 - the grammar is ambiguous
 - the grammar is unambiguous
 - We do not have enough information to decide if the grammar is ambiguous or not
 - None of the above
- Let $G_2 = (\{S, A, B\}, \{0, 1\}, S, P)$ a CFG, where P is:
 $S \rightarrow BB$
 $A \rightarrow AA \mid 0$
 $B \rightarrow AA \mid 1$
 Then, $L(G_2)$
 - is empty
 - is finite but not empty
 - is infinite
 - None of the above.

1. Regístrate o inicia sesión como miembro de ASUS
 2. Completa el registro en Sheer ID y espera la confirmación por email
 3. ¡Espera al 6 de abril! Elegiremos a 3 ganadores y, si eres uno de los afortunados, te contactaremos por email.

*Consulta términos y condiciones. Sorteo válido del 14 de marzo al 6 de abril.



Participa aquí

6. Let $G_3 = (V, T, S, P)$ be a context-free grammar where $V = \{S, A, B, C, D, E\}$, $T = \{a, b, c\}$, and P :
- $$\begin{aligned} S &\rightarrow aBC \\ A &\rightarrow Aa \mid B \mid BC \\ B &\rightarrow b \mid AA \\ C &\rightarrow A \mid B \\ D &\rightarrow EAD \mid CD \mid E \\ E &\rightarrow cAD \mid AD \mid \lambda \end{aligned}$$
- The following are the nullable symbols:
- D
 - E and D
 - E
 - None of the above.
7. Remove λ -productions from the original grammar G_3 using the algorithm learnt in class. Then, in the new grammar
- the productions for D are: $D \rightarrow EAD \mid CD \mid E$
 - the productions for D are: $D \rightarrow EAD \mid CD \mid E \mid AD \mid EA \mid A \mid C$
 - the productions for D are: $D \rightarrow EAD \mid CD \mid E \mid AD \mid EA \mid A \mid C \mid \lambda$
 - None of the above.
8. Consider the original grammar G_3 . Eliminate unit productions using the algorithm learnt in class. Then, in the new grammar
- the productions for C are: $C \rightarrow Aa \mid B \mid BC \mid b \mid AA$
 - the productions for C are: $A \rightarrow Aa \mid BC \mid AA$
 - the productions for C are: $A \rightarrow Aa \mid BC \mid b \mid AA$
 - None of the above.
9. Consider the original grammar G_3 . The set of non-generating symbols is:
- D and E
 - A and S
 - A, E and S
 - None of the above.
10. Consider the original grammar G_3 . The set of unreachable symbols is:
- D
 - D and E
 - E, c and D
 - None of the above.